Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 09)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A student is comparing the plant diversity of a grazed and an ungrazed grassland, and estimating the population of grasshoppers in each. In the grazed field 60 grasshoppers were caught, marked and released; three days later 75 were caught, of which 15 were marked.

0 1 . 1 Explain **three** reasons for choosing quadrat positions using random numbers rather than by eye.

[3 marks]

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0 1 . 2 Use the Lincoln index to estimate the grasshopper population of the grazed field.

[2 marks]

Answer

0 1 . 3 State **three** assumptions that must hold for this estimate to be valid.

[3 marks]

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0 1 . 4 Explain how the student should decide the size of quadrat and the number of quadrats to use.

[3 marks]

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0 1 . 5 Explain **four** other features of the design that would make the comparison between the two fields more reliable or more valid.

[4 marks]

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0 2 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 2 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 2 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

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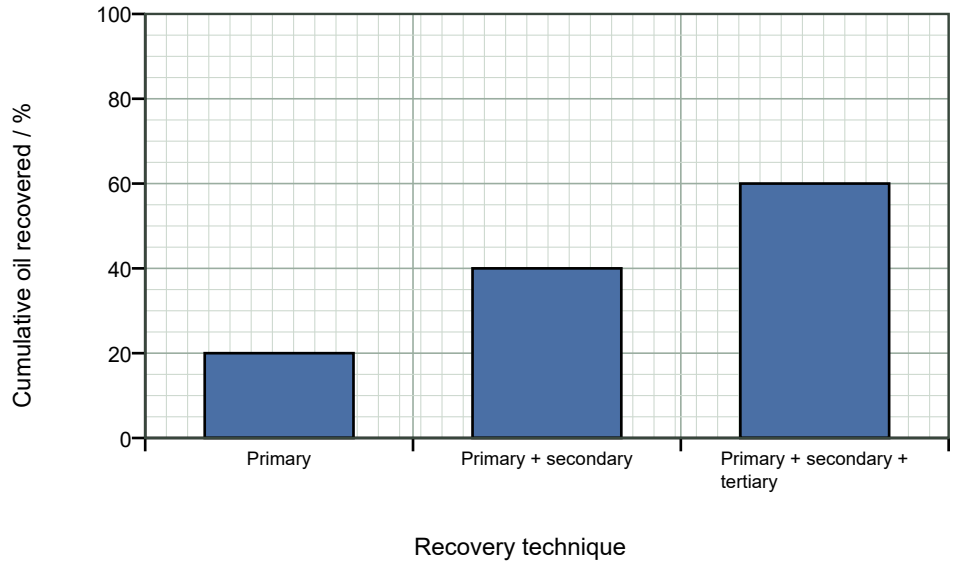
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0 3 Figure 1 shows the percentage of the oil in a reservoir that can be recovered using different recovery techniques.

Figure 1



0 3 . 1 Describe how primary oil recovery works and explain why it recovers only about 20% of the oil.

[2 marks]

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0 3 . 2 Describe secondary recovery and explain how it can be linked to carbon capture and storage.

[2 marks]

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0 3 . 3 Describe **three** tertiary (enhanced) recovery techniques.

[3 marks]

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0 3 . 4 Explain **three** advantages of directional drilling.

[3 marks]

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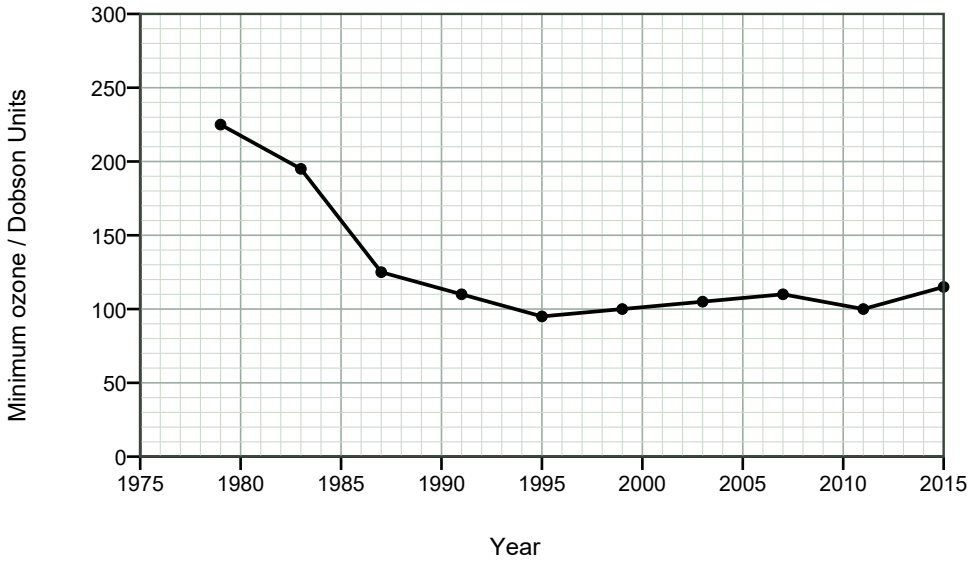
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0 4 Figure 1 shows the minimum ozone concentration recorded over Antarctica each spring between 1979 and 2015.

Figure 1



0 4 . 1 Describe the trend shown in Figure 1.

[2 marks]

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0 4 . 2 Calculate the percentage decrease in minimum ozone between 1979 and 1995.

[2 marks]

Answer ..... %

0 4 . 3 Explain why ozone depletion is most severe over Antarctica in spring.

[4 marks]

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0 4 . 4 Explain why the data in Figure 1 alone could not confirm that CFCs were the cause of the depletion, and describe how the cause was confirmed.

[2 marks]

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0 5 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

0 5 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 6 . 1 Explain why nitrates in drinking water can cause **blue baby syndrome** (methaemoglobinaemia).

[3 marks]

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0 6 . 2 Explain why the link between nitrates in drinking water and stomach cancer is uncertain.

[2 marks]

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0 6 . 3 Describe **three** ways farmers in a Nitrate Vulnerable Zone can reduce nitrate leaching into aquifers.

[3 marks]

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0 6 . 4 Explain how phosphates are removed from sewage effluent and from lakes polluted in the past.

[2 marks]

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0 7 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 7 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 7 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 8 . 1 Explain how energy absorbed in tropical regions is distributed to higher latitudes such as the UK.

[4 marks]

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0 8 . 2 Explain why the UK has milder winters than Moscow or Churchill, Canada, which are at similar latitudes.

[2 marks]

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0 8 . 3 Explain why the atmosphere would not retain liquid water if atmospheric pressure were much lower.

[2 marks]

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0 8 . 4 Give **two** industrially important gases extracted from the atmosphere.

[2 marks]

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0 9 A dam and reservoir are to be built on a river to supply water and regulate flow.

0 9 . 1 Explain **three** effects the reservoir may have on wildlife in and around the river.

[6 marks]

0 9 . 2 Explain why decomposition of flooded vegetation in a new reservoir contributes to climate change.

[2 marks]

0 9 . 3 Explain how a river-regulation reservoir reduces both flooding and water shortages downstream.

[2 marks]

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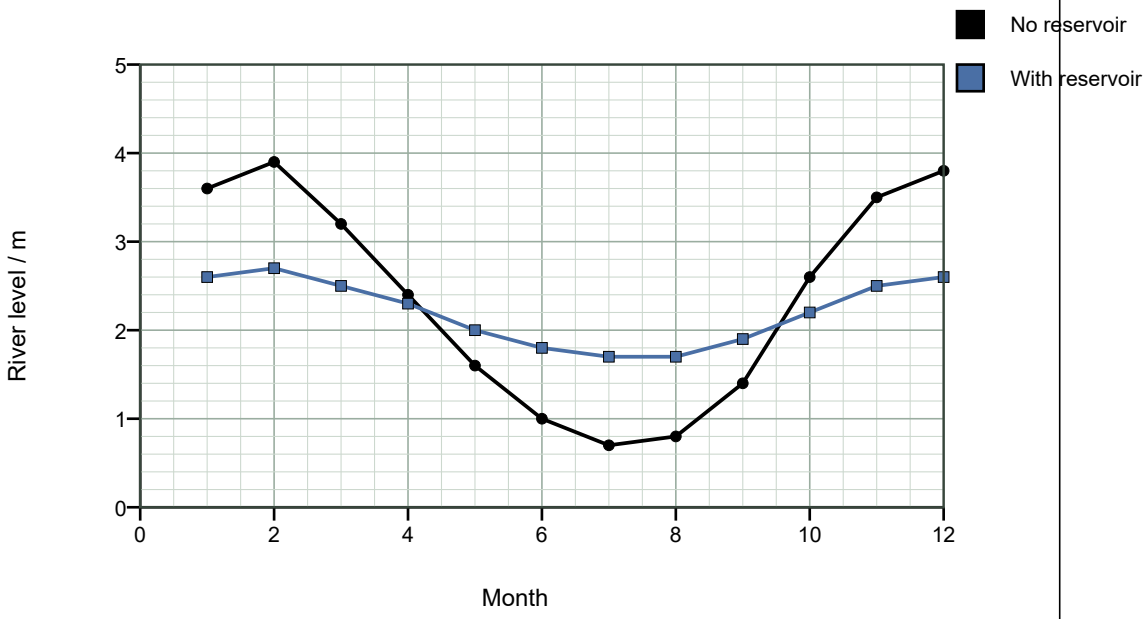
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1 0 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer ..... m

1 0 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

1 0 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

OR

1 1 . 2 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS